↳ Special Elements in Binary Operations: The Identity Element

- **Definition 9.5:** Let $\circ$ be a binary operation on S.

(1) If there exists an element e_l (or e_r) $\in S$, such that for all $x \in S$

$$e_l \circ x = x \quad (\text{ or } \quad x \circ e_r = x),$$

then e_l (or e_r) is called the ***left identity element*** (or ***right identity element***) of S with respect to the operation $\circ$.

(2) If $e \in S$ is both a left identity and a right identity under $\circ$, then e is called the ***identity element*** (or **unit element**) of S with respect to the operation $\circ$.

- **Note:**

An identity (or unit) element is a ***special element*** under the given operation $\circ$ that does **not change other elements** when the operation is applied.

- **Definition 9.6:** Let $\circ$ be a binary operation on $\boldsymbol{S}$.

  (1) If there exists an element $\boldsymbol{\theta_l}$ (**or** $\boldsymbol{\theta_r}$)$\in\boldsymbol{S}$, such that for all $\boldsymbol{x}\in\boldsymbol{S}$ we have $\boldsymbol{\theta_l \circ x = \theta_l}$ (**or** $\boldsymbol{x \circ \theta_r = \theta_r}$), then $\boldsymbol{\theta_l}$ (**or** $\boldsymbol{\theta_r}$) is called the ***left zero element*** (or ***right zero element***) of $\boldsymbol{S}$ with respect to the operation $\circ$.

  (2) If an element $\boldsymbol{\theta} \in \boldsymbol{S}$ is both a left zero element and a right zero element under $\circ$, then $\boldsymbol{\theta}$ is called the ***zero element*** of $\boldsymbol{S}$ with respect to the operation $\circ$.

- **Note:** A zero element is a ***special element*** that ***"absorbs" other elements*** when combined with them under a specific binary operation, always producing itself as the result.

- **Definition 9.7:** Let e be the identity (unit) element of S with respect to the operation $\circ$.

(1) For any $x \in S$, if there exists y_l (or y_r) $\in S$ such that

$$y_l \circ x = e \quad (\text{or } x \circ y_r = e),$$

then y_l (or y_r) is called the *left inverse* (or *right inverse*) of x.

(2) If $y \in S$ is both the left inverse and the right inverse of x, then y is called the *inverse* of x.

(3) If the inverse of x exists, then x is said *to be invertible*.

Special Elements in Binary Operations(e.g.)

Set	Operations	identity element	Zero element	Inverse
Z, Q,R	$+$	0	NO	$-x$
	$\times$	1	0	x^{-1}
M_n(R)	$+$	θ	NO	$-X$
	$\times$	E	θ	X^{-1} （X is an invertible matrix）
$P(B)$	$\cup$	$\varnothing$	B	Only when $\varnothing$ has inverse
	$\cap$	B	$\varnothing$	*Only when B has inverse*
	$\oplus$	$\varnothing$	NO	X

- **Membership of the *Multiplicative Inverse* x^{-1}:**
  - In the set of integers **Z**, *only 1 and −1 have multiplicative inverses within Z*, the multiplicative inverses of all other integers do **not** belong to **Z**.
  - In the set of rational numbers **Q**, *all nonzero elements have multiplicative inverses*, and these inverses also belong to **Q**.
  - In the set of real numbers **R**, *all nonzero elements have multiplicative inverses*, and these inverses also belong to **R**.

↳ Uniqueness of the Identity Element Theorem

- **Theorem 9.1:** Let $\circ$ be a binary operation on S. If e_l and e_r are the left identity and right identity elements of S with respect to $\circ$, respectively, then we have $e_l = e_r = e$, and e is the *unique identity element* of S under $\circ$.
- **Proof:**

  ①By the definition of the left identity, we have $e_l \cdot e_r = e_r$. By the definition of the right identity, we have $e_l \cdot e_r = e_l$.

  ②Combining these two equations, we get $e_l = e_l \cdot e_r = e_r$.

  ③We denote $e_l = e_r$ as e. Now, suppose e' is also an identity element of S, then $e' = e \circ e' = e$.

  ④Therefore, the identity element e is unique.
- Similarly, one can prove the *uniqueness theorem for the zero element*.

- **Theorem 9.2:** Let $\circ$ be an associative binary operation on S, and let e be the identity element under this operation. For any $x \in S$, if there exists a left inverse y_l and a right inverse y_r, then $y_l = y_r = y$, and y is the *unique inverse* of x.
- **Proof:**

① Since $y_l \circ x = e$ and $x \circ y_r = e$,we have $y_l = y_l \circ e = y_l \circ (x \circ y_r)$
$= (y_l \circ x) \circ y_r = e \circ y_r = y_r$.

② Let $y_l = y_r = y$, then y is an inverse of x.

③ Now, suppose $y' \in S$ also an inverse of x, then $y' = y' \circ e = y' \circ (x \circ y)$
$= (y' \circ x) \circ y = e \circ y = y$.

④ Therefore, y s the unique inverse of x.

- **Note:** For *associative binary operations*, an invertible element x has a *unique inverse*, denoted as x^{-1}.

- **Definition 9.8:** Let $\circ$ be a binary operation on V, if $\forall\, x, y, z \in V$,
  If $x \circ y = x \circ z$, and x is not a zero element, then $y = z$ (Left cancellation).
  If $y \circ x = z \circ x$, and x is not a zero element, then $y = z$ (Right cancellation).
  then the operation $\circ$ is said to satisfy the ***cancellation law*** .
- **Examples:**
  (1) Z, Q, R satisfy the cancellation law under ordinary addition and multiplication.
  (2) M_n(R) satisfies the cancellation law under matrix addition, but does not satisfy the cancellation law under matrix multiplication.
  (3) Z_n satisfies the cancellation law under addition modulo n, it also satisfies the cancellation law under multiplication modulo n when n is prime. However, when n is composite, multiplication modulo n does not satisfy the cancellation law.
  (4) Set union and intersection do not satisfy the cancellation law. For example, $\{1\}\cup\{1,2\}=\{2\}\cup\{1,2\}$, but $\{1\}\neq\{2\}$.

- **Example:** Let $\circ$ be a binary operation on **Q**, defined by,

$$\forall x, y \in Q, \quad x \circ y = x+y+2xy,$$

(1) Determine whether the operation $\circ$ satisfies the commutative law and the associative law, and explain why.

(2) Find the identity element, the zero element, and the inverse elements (for all invertible elements) under the operation .

- **Solution (1):**

① The $\circ$ operation is commutative. For any $x, y \in Q$,

$$x \circ y = x+y+2xy = y+x+2yx = y \circ x .$$

② The $\circ$ operation is associative. For any $x, y, z \in Q$,

$$(x \circ y) \circ z = (x+y+2xy) + z + 2(x+y+2xy)\, z$$
$$= x+y+z+2xy+2xz+2yz+4xyz$$
$$x \circ (y \circ z) = x + (y+z+2yz) + 2x(y+z+2yz)$$
$$= x+y+z+2xy+2xz+2yz+4xyz$$

■ Solution (2):

- Find the identity element e :

  ① Assume that the identity element and the zero element under the $\circ$ operation are e and θ, respectively. For any x, we have $x\circ e = x$, then $x+e+2xe = x \Rightarrow e = 0$. ② Since the $\circ$ operation is commutative, **0** is the identity element (unit element).

- Find the zero element θ :

  ① For any x, we have $x \circ \theta = \theta \circ x = \theta$, that is

  $x+\theta+2x\theta = \theta \Rightarrow x+2x\theta = 0 \Rightarrow \theta = -1/2$. ② $-1/2$ zero element.

- Find the inverse y :

  ① Let y be the inverse of x. Then we have $x \circ y = e = 0$,that is $x+y+2xy = 0$.

  ② $y = -\frac{x}{2x+1}$, this means that every x （$x \neq -\frac{1}{2}$ ） has an inverse **y**.

- **Example:** Here are three operation tables:
  (1) Identify which operations are **commutative**, **associative**, and **idempotent**.
  (2) Find the **identity element**, **zero element**, and the **inverse** of all invertible elements for each operation.

$*$	a	b	c
a	c	a	b
b	a	b	c
c	b	c	a

$\circ$	a	b	c
a	a	a	a
b	b	b	b
c	c	c	c

$\bullet$	a	b	c
a	a	b	c
b	b	c	c
c	c	c	c

- **Solve (1) :** ① $*$ satisfies the commutative law and associative law. ② $\circ$ satisfies the associative law and idempotent law. ③ $\bullet$ satisfies the **commutative law** and **associative law**.
- **Solve (2) :** ① $*$ the identity element is b, no zero element, $a^{-1}=c$, $b^{-1}=b$, $c^{-1}=a$. ② $\circ$ there is no identity element and no zero element, there are no invertible elements. ③ $\bullet$ the identity element is a, the zero element is c, $a^{-1}=a$. b, c are not invertible.

Objective :

Key Concepts :

Discrete Mathematics 2025 Spring

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CAMEA 中国高质量MBA教育认证 AACSB ACCREDITED EQUIS ACCREDITED

- 9.1 Binary Operations and Their Properties
- 9.2 Algebraic Systems
- 9.3 Several Typical Algebraic Systems

- 9.2.1 Definition and Examples of Algebraic Systems
- 9.2.2 Subalgebraic Systems and Product Algebraic Systems
- 9.2.3 Homomorphisms and Isomorphisms of Algebraic Systems

↳ The subject matter of algebra and its process of abstraction

- ***Algebra*** (in the broad sense), as a major branch of mathematics, studies various algebraic structures as well as related operations, equations, and theories. The objects of algebra are not limited to numbers, but extend to all kinds of abstract structures.
- The development of algebra is a gradual process of ***abstraction and systematization*** — from the basic concepts in elementary algebra (such as addition, multiplication, variables, and polynomials), to linear algebra (vectors, matrices, and determinants), and then to the more abstract structures of higher algebra (such as groups, rings, and fields). It represents humanity's continuous pursuit and exploration of the unknown.

↳ Algebraic structures and algebraic systems

- An *algebraic structure* (or type of algebraic system) is an **abstract mathematical concept** used to describe operations defined on one or more sets, along with their (abstract) properties. These operations must satisfy specific rules and axioms. Examples include algebraic structures such as groups, rings, and fields.
- An *algebraic system* is a concrete realization or **instance of an algebraic structure**, consisting of a specific set and concrete operations defined on that set.
  - **For example:**
    - Addition on the set of *natural numbers* $\mathbb{N}$.
    - Addition and multiplication on the *set of integers* $\mathbb{Z}$.
    - Addition and multiplication on the *set of real numbers* $\mathbb{R}$.
    - Addition and multiplication on the set of ***n*-order** ($n \geq 2$) *real matrices* $\mathbf{M_n(\mathbb{R})}$.
    - Addition and multiplication modulo n on *the set* $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$.
    - Union, intersection, and absolute complement on the *power set* $\mathcal{P}(S)$ of a set S.

- **Definition 9.9:** A non-empty set S together with k operations f_1, f_2, … , f_k （where each f_i n_i-**ary** operation，i=1,2,…,k ）forms a system called an ***algebraic system*** (or simply, an **algebra**), denoted as **<S, f_1, f_2, …, f_k>**.
- **Examples:**
  - <N, + >,<Z, +,·>,<R, +,·> are algebraic systems, where + and · represent the usual addition and multiplication.
  - <M_n(R), +,·> is an algebraic system, where + and · denote addition and multiplication of $n \times n$ ($n \geq 2$) real matrices.
  - <Z_n,⊕,⊗>is an algebraic system, Z_n={ 0, 1, … , n–1 }, ⊕ and ⊗ denote addition and multiplication modulo n，for x, $y \in Z_n$, $x \oplus y=(x+y) \bmod n$，$x \otimes y=(xy) \bmod n$.
  - <$P(S)$,∪,∩,~>is also an algebraic system,∪ and ∩ are union and intersection, and ~ is the absolute complement.

↳ The constituent elements of an algebraic system

(1) A ***basic set*** **(carrier set)** that contains all the elements. For example, in the ring of integers, the set is the set of all integers **Z** .

(2) *Operations* defined on the set. Common operations include unary and binary operations.

(3) Some algebraic systems require the definition of certain ***special elements or algebraic constants*** (such as the **zero element** or **identity element** for a binary operation). For example, the element **0** in integer addition, or the element **1** in integer multiplication.

(4) Certain operations may require the existence of ***inverse elements***. For example, for an integer n, its additive inverse is $-n$.

(5) An algebraic system typically comes with a set of ***axioms*** that must be satisfied, such as the associative law, commutative law, and distributive law.

- **Note:** In addition to these basic components, different algebraic structures (such as *groups*, *rings*, *fields*, *vector spaces*) also have their own ***specific components and axioms*** that must be satisfied.
- **Example:**
  - $<Z, +>$ has the identity element **0**, and can also be written as $<Z, +, 0>$.
  - $<P(S), \cup, \cap, \sim>$ the identity elements for $\cup$ and $\cap$ are the empty set $\varnothing$ and the universal set S, respectively. It can also be written as $<P(S), \cup, \cap, \sim, \varnothing, S>$.

- 9.2.1 Definition and Examples of Algebraic Systems
- 9.2.2 Sub-algebraic Systems and Product Algebraic Systems
- 9.2.3 Homomorphisms and Isomorphisms of Algebraic Systems

- **Definition 9.10:** Let V=<S, f_1, f_2, … , f_k> be an algebraic system, and let B be a non-empty subset of S. If B is closed under all operations f_1, f_2, … , f_k, B and S share the same algebraic constants, then <B, f_1, f_2, … , f_k> is called a ***sub-algebraic system*** (or simply, a **subalgebra**) of V. Sometimes, the sub-algebraic system is simply denoted by B.
- **Example:**
  - < N,＋> is a subalgebra of <Z,＋>.
  - < N,＋ ,0 > is also a subalgebra of <Z,＋,0> （because N is closed under + and has the same algebraic constants）.

- **Example:**

  - $<N-\{0\},+>$ is a subalgebra of $<Z,+>$, but not a subalgebra of $<Z,+,0>$ (because the algebraic constant **0** is not included in $N-\{0\}$).

- **Notes:**

① A subalgebra and its original algebra are the ***same type of algebraic system*** (they share the same algebraic constants, the same number of operations, and the same operational properties).

② Every algebraic system V always has ***at least one subalgebraic system***.

↳ Trivial subalgebra and proper subalgebra

- The ***largest subalgebra*** is simply ***V*** itself.
- The ***smallest subalgebra*** is the set ***B*** formed by all the algebraic constants in ***V***, provided that ***B*** is closed under all operations in ***V***, in this case, ***B*** constitutes the smallest subalgebra of ***V***.
- The ***trivial subalgebras*** refer to the largest and smallest subalgebras of ***V***.
- A ***proper subalgebra*** refers to a subalgebra ***B*** where **B** is a proper subset of ***S***, that is, ***B*** forms a proper subalgebra of ***V***.
- **Example:** Let V=<Z,+,0>, and define $nZ = \{ nz \mid z \in Z\}$, n is a natural number, then nZ is a subalgebra of V, when n = **1 or 0**, nZ is a trivial subalgebra of V, for all other n, nZ is a nontrivial proper subalgebra of V.

- **Definition 9.11:** Let $V_1=<S_1,\circ>$ and $V_2=<S_2,*>$ be algebraic systems, $\circ$ and $*$ are binary operations. The ***product algebra*** $V_1 \times V_2$ is an algebraic system with a binary operation, defined as $V_1\times V_2 =<S, \bullet>$, where $S =S_1\times S_2$, and for all $\forall<x_1, y_1>, <x_2, y_2>\in S_1\times S_2$, we have $<x_1, y_1> \bullet <x_2, y_2>=<x_1\circ x_2, y_1 * y_2>$.
- **Example:** Consider integer addition $V_1=<Z,+>$ and matrix multiplication $V_2=<M_2(R), \cdot >$,

  then $V_1 \times V_2 = < Z\times M_2(R), \circ>$,

  and for all $\forall<z_1,M_1>, <z_2,M_2>\in Z\times M_2(R)$,

  we have $<z_1,M_1> \circ <z_2,M_2> = <z_1+z_2, M_1\cdot M_2>$.

For example: $<5,\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}> \circ <-2,\begin{pmatrix}2 & -1\\ 0 & 1\end{pmatrix}> = <3,\begin{pmatrix}2 & -1\\ 2 & 0\end{pmatrix}>$

↳ The properties of product algebras

- Let $V_1=<S_1,\circ>$ and $V_2=<S_2,*>$ be algebraic systems, where $\circ$ and $*$ are binary operations. The product algebra is $V=<S_1\times S_2, \bullet>$.

(1) If $\circ$ and $*$ are **commutative**, then the operation $\bullet$ is also ***commutative***.

(2) If $\circ$ and $*$ are **associative**, then the operation $\bullet$ is also ***associative***.

(3) If $\circ$ and $*$ are **idempotent**, then the operation $\bullet$ is also ***idempotent***.

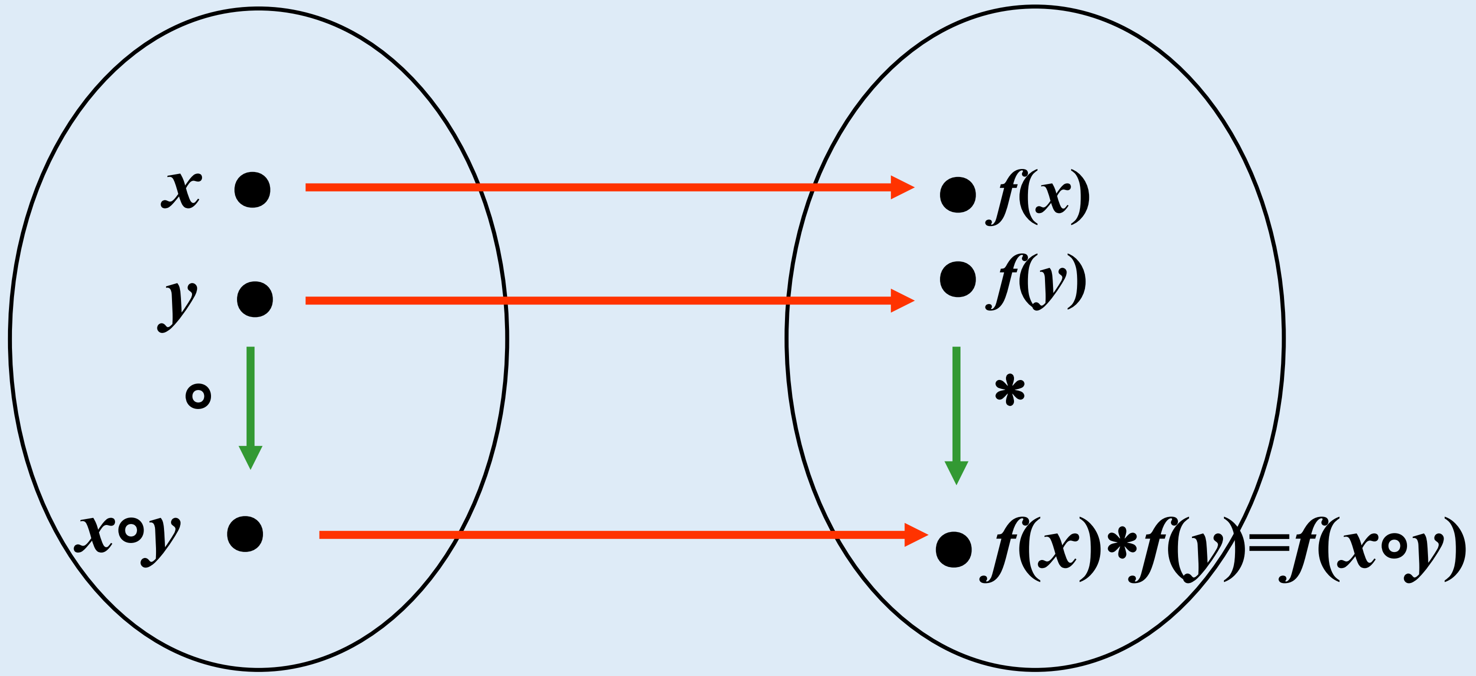
(4) If $\circ$ and $*$ have respective **identity elements** e_1 and e_2 then the operation $\bullet$ also has an ***identity elements*** $<e_1,e_2>$.

(5) If $\circ$ and $*$ have respective **zero elements** θ_1 and θ_2, then the operation $\bullet$ also has a ***zero element*** $<\theta_1, \theta_2>$.

(6) If x has an **inverse** x^{-1}, with respect to $\circ$, and y has an **inverse** y^{-1} with respect to $*$,then $\langle x,y\rangle$ has an ***inverse*** $<x^{-1},y^{-1}>$ with respect to $\bullet$.

- Definition of Homomorphism
- Classification of Homomorphisms
  - Monomorphism, Epimorphism, Isomorphism
  - Endomorphism
- Examples of Homomorphisms
- Properties of Epimorphism

- **Definition 9.12:** Let V_1=<S_1,∘ > and V_2=<S_2,∗> be algebraic systems, where ∘ and ∗ are binary operations. If there exists a mapping f : $S_1 \to S_2$, such that $\forall x,y \in S_1$, $f(x \circ y) = f(x) * f(y)$ then f is called a *homomorphism* from V_1 to V_2, or simply a **homomorphism.**



- **Example :** V=<R*,·>, determine which of the following functions are homomorphisms of V?

(1) $f(x)=|x|$ (2) $f(x)=2x$ (3) $f(x)=x^2$
(4) $f(x)=1/x$ (5) $f(x)=-x$ (6) $f(x)=x+1$

- **Solution:** Analyze whether the function satisfies $f(x\cdot y)=f(x)\cdot f(y)$ for all x,y in the nonzero real numbers set **R*** .

(1) $f(x\cdot y) = |x\cdot y| = |x| \cdot |y| = f(x) \cdot f(y)$ homomorphisms
(2) $f(x\cdot y)=2(x\cdot y) \neq f(x)\cdot f(y)=(2x)\cdot(2y)=4xy$ non-homomorphic
(3) $f(x\cdot y)=(x\cdot y)^2=f(x)\cdot f(y)=x^2\cdot y^2$ homomorphisms
(4) $f(x\cdot y)= 1/(x\cdot y)=f(x)\cdot f(y)= 1/x\cdot 1/y$ homomorphisms
(5) $f(x\cdot y)=-(x\cdot y) \neq f(x)\cdot f(y)=(-x)\cdot(-y)=xy$ non-homomorphic
(6) $f(x\cdot y)=x\cdot y+1\neq f(x)\cdot f(y)\ (x+1)\cdot(y+1)=xy+x+y+1$ non-homomorphic

- Let f be a *homomorphism* from $V_1=<S_1,\circ>$ to $V_2=<S_2,*>$, if f is injective (one-to-one), then f is called a *monomorphism*.
  - **Note:** f being injective (or a one-to-one mapping) means that for any $x_1,x_2 \in S_1$, if $f(x_1)=f(x_2)$, then $x_1=x_2$.
- **Example:** Let $V_1 = <R^*,\cdot>$, $V_2 = <R^*,\cdot>$, and define the mapping f:$R^* \rightarrow R^*$ 为 $f(x)=x^2$. Is f a monomorphism ?
  - We need to verify the homomorphism property and injectivity.
  - **Homomorphism:** We need to check whether for all $x,y \in R^*$, the equation $f(x\cdot y)=f(x)\cdot f(y)$ holds. Since $f(x\cdot y)= (x\cdot y)^2 = f(x)\cdot f(y) = x^2 \cdot y^2$, **thus** $f(x)=x^2$ satisfies the *homomorphism* property .
  - **Injectivity:** We need to check whether for all $x_1, x_2 \in R^*$, if $f(x_1)=f(x_2)$, then $x_1=x_2$. For example, if $x_1=-1$ 、$x_2=1$, we have $x_1^2= x_2^2 =1$, **but** $x_1 \neq x_2$, Therefore, the mapping $f(x)=x^2$ is *not injective* .
  - **Conclusion:** $f(x)=x^2$ is a homomorphism, but it is **not** a monomorphism.

- Let f be a homomorphism from V_1=<S_1,∘ > to V_2=<S_2,∗>, if f is surjective (onto) , then f is called an ***epimorphism***. In this case, V_2 is called the **homomorphic image** of V_1, denoted $V_1 \overset{f}{\sim} V_2$.
  - **Note**: f being surjective (or an onto mapping) means that it is covering, i.e., for every element ***y*** in S_2 , there exists some element ***x*** in S_1 such that ***f*(*x*)=*y*.**
- **Example:** Let $V_1 = \langle Z,+\rangle$, $V_2=\langle Z_3,+_3\rangle$ (the set of integers modulo 3 with addition modulo 3), and define the mapping ***f***: Z → Z_3 be defined by ***f*(*x*)= *x* mod 3.** Is f an epimorphism ?

- *Homomorphism*: We need to check whether for all $x,y \in Z$, $f(x+y)=f(x) +_3 f(y)$. Since $f(x+y)=(x+y) \bmod 3$, $f(x) +_3 f(y)=(x \bmod 3 + y \bmod 3) \bmod 3$, by the distributive property of modular arithmetic, we have $f(x+y)=f(x) +_3 f(y)$, Therefore $f(x)= x \bmod 3$ satisfies the homomorphism property.
- *Surjectivity*: For each element ***y in*** Z_3, does there exist some ***x in*** Z_3 such that $f(x)=y$. Since every element in Z_3 (0, 1, 2) can be obtained by taking modulo 3 of some integer, f is surjective. That is, f is an epimorphism.
- *Conclusion*: $f(x)= x \bmod 3$ is a **homomorphism** and is indeed an **epimorphism.**
- **Note**: Since x_1 and x_2 having the same remainder under $f(x)= x \bmod 3$ does not imply that $x_1 = x_2$, f is not injective, that is, f is *not a monomorphism*.

- Let f be a homomorphism from V_1=<S_1,∘ > to V_2=<S_2,∗>, if f is bijective, then f is called an ***isomorphism*** from V_1 to V_2 denoted as $V_1 \cong V_2$.

- **Note**: f being bijective (or a one-to-one and onto mapping) means that f is both injective (one-to-one) and surjective (onto). The existence of an isomorphism between two algebraic systems means that their algebraic structures are equivalent.

- **Example:** Let $V_1=\langle R,+\rangle$, $V_2=\langle R,+\rangle$, f: R→R be defined by $f(x)=2x$. Is the mapping f an isomorphism from V_1 to V_2?
  - we need to verify two conditions: the mapping is a homomorphism, and it is bijective.

- **Homomorphism:** $f(x+y)=2(x+y)=2x+2y=f(x)+f(y)$, which shows that f is a homomorphism.
- **Bijectivity :**
  - **Injectivity:** If $f(x_1)=f(x_2)$, **then** $2x_1 = 2x_2$, which leads to $x_1 = x_2$, showing that f is injective.
  - **Surjectivity:** For every real number **y∈R**, there exists $x=y/2$, such that $f(x)=2 \cdot y/2=y$ showing that f is surjective.
  - Since f is both injective and surjective, it is bijective.
- **Conclusion:**

  $f(x)=2x$ is an *isomorphism* from $V1$ to $V2$, and it preserves the properties of the addition operation (closure, commutativity, associativity, identity element, and inverse element).

↳ Endomorphisms and automorphisms of algebraic systems

- A homomorphism $f:S\to S$ from an algebraic $V=\langle S,\circ\rangle$ to itself is called an ***endomorphism***. That is, for all $x,y\in S$ we have $f(x\circ y)=f(x)\circ f(y)$.
- A ***zero homomorphism*** maps every input element to the zero element in the target algebraic structure.
- An ***automorphism*** refers to a homomorphism $f:S\to S$ of the algebraic system $V=\langle S,\circ\rangle$ that is both injective and surjective, indicating that the system is structurally equivalent to itself.
- A ***monomorphic endomorphism*** (or **injective endomorphism**) is a special type of endomorphism that is also injective.

- **Example: let V=<Z,+>, $\forall a \in Z$, f_a : Z→Z,** $f_a(x)=ax$. Prove that f_a is an endomorphism of V.
  - Since $\forall x,y \in Z$, we have: $f_a(x+y) = a(x+y) = ax+ay = f_a(x)+f_a(y)$.
  - When $a = 0$, f_0 is called the ***zero homomorphism***, **when** $a=\pm 1$, f_a is called an ***automorphism***.
  - For all other values of a, f_a is an ***injective endomorphism*** (monomorphic endomorphism).

- **Example:** Let V_1=<Q,+>, V_2=<Q*,·>, where Q*=Q–{0} is the set of nonzero rational numbers. Define f : Q→Q*, $f(x)=e^x$.
  Determine the type of homomorphism that $f(x)$ defines from V_1 to V_2 .
- **Solution:**
  - f is a *homomorphism* from V_1 to V_2, because $\forall x,y \in Q$, have $f(x+y)=e^{x+y}=e^x \cdot e^y=f(x) \cdot f(y)$.
  - f is a *monomorphism* because for $\forall x,y \in Q$, If $f(x_1)=f(x_2)= e^{x1}=e^{x2}$, which implies $x_1=x_2$, f is *injective*.
  - However, for any $y \in Q*$, e^x cannot reach every nonzero rational number (for example, negative numbers), so $f(x)=e^x$ is *not surjective*.

- **Example:** Let $V=<Z_n,\oplus>$, $f_p:Z_n\to Z_n$, $f_p(x) = (xp) \bmod n$, $p = 0, 1, \dots ,$ $n-1$. Analyze the properties of f_p.

① ***Homomorphism***: $\forall x, y\in Z_n$, $f_p(x\oplus y)=((x\oplus y)p) \bmod n = (xp) \bmod n \oplus (yp) \bmod n = f_p(x) \oplus f_p(y)$.

② f_p is an ***endomorphism***. Both the input and output of f_p are Z_n , and it satisfies the homomorphism property.

③ If *p and n are coprime*, then f_p is ***injective*** **(Monomorphism)**. For example, when $n=6$ *and* $p=3$(which are not coprime), we have $f_p(1)=(1\cdot 3)$ *mod* $6 =3$, but $x_1=1 \neq x_2 =3$, so it is not injective.

- **Example:** Let V=<Z_n,⊕>, f_p:$Z_n \to Z_n$, $f_p(x)$ = (xp) mod n, p = 0, 1, … , n–1. Analyze the properties of f_p.

④ **Surjectivity (*Epimorphism*):** If ***p* and *n* are coprime**, then f_p is surjective. For example, with ***n*=6** and ***p*=3** (not coprime), in Z_6 , for y=0, there is ***x*=0,2,4**,for ***y*=3**, there is ***x*=1,3,5**,but for ***y*=1,2,4,5**, no ***x*** satisfies $f_p(x)=y$.

⑤ **Zero Homomorphism:** If $p=0$, then $f_p(x) =(x \cdot 0) \bmod n=0$, which means it maps all inputs to zero. Therefore, f_p is the ***zero homomorphism***.

⑥ **Automorphism:** If ***p* and *n* are coprime**, f_p satisfies both injectivity and surjectivity (bijectivity), so f_p is an ***automorphism***.

Properties of epimorphism between Algebraic Systems

- Let V_1 and V_2 be algebraic systems, and let $f : V_1 \to V_2$, be a **surjective (onto) homomorphism**. Then:

  (1) If the operation $\circ$ in V_1 is commutative (associative, idempotent), then the corresponding operation $\circ'$ in V_2 is also ***commutative*** (associative, idempotent).

  (2) If $\circ$ is distributive over $*$ in V_1, then the corresponding $\circ'$ is ***distributive*** over $*'$ in V_2.

  (3) If $\circ$ and $*$ are absorbent (absorption law holds) in V_1, then the corresponding $\circ'$ and $*'$ are also ***absorbent*** in V_2.

  (4) If $\circ$ in V_1 has an identity element e_1 (or a zero element θ_1), then $f(e_1)$ (or $f(\theta_1)$) is the ***identity (or zero) element*** for the corresponding operation $\circ'$ in V_2.

  (5) If the inverse of x under $\circ$ in V_1 is x^{-1}, then the ***inverse*** of $f(x)$ under the corresponding $\circ'$ in V_2 is $f(x^{-1})$.

Objective :

Key Concepts :

Discrete Mathematics 2025 Spring

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- 9.1 Binary Operations and Their Properties
- 9.2 Algebraic Systems
- 9.3 Several Typical Algebraic Systems